

23CSE314 Compiler Design

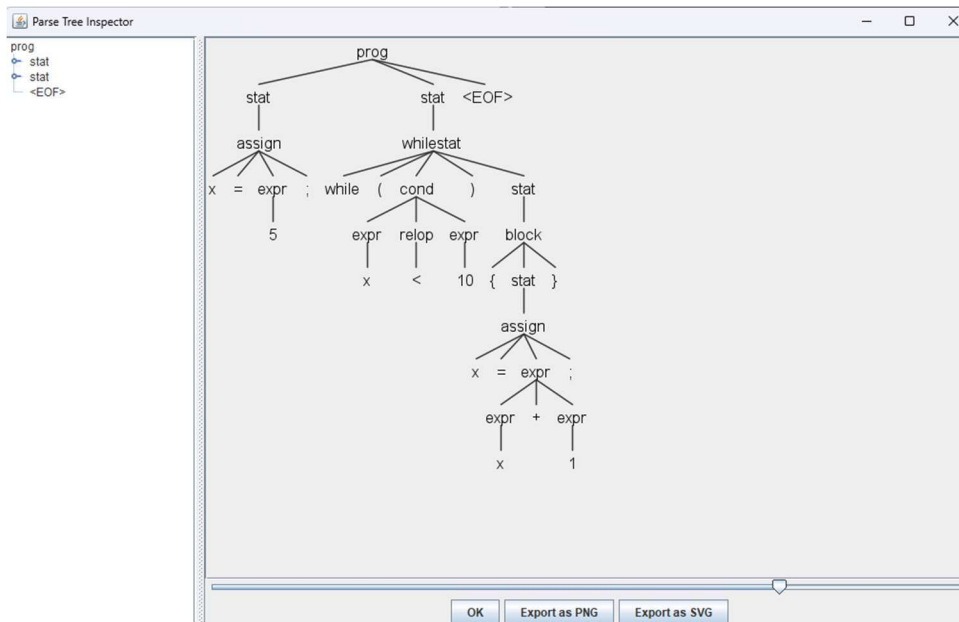
Lab Sheet 5

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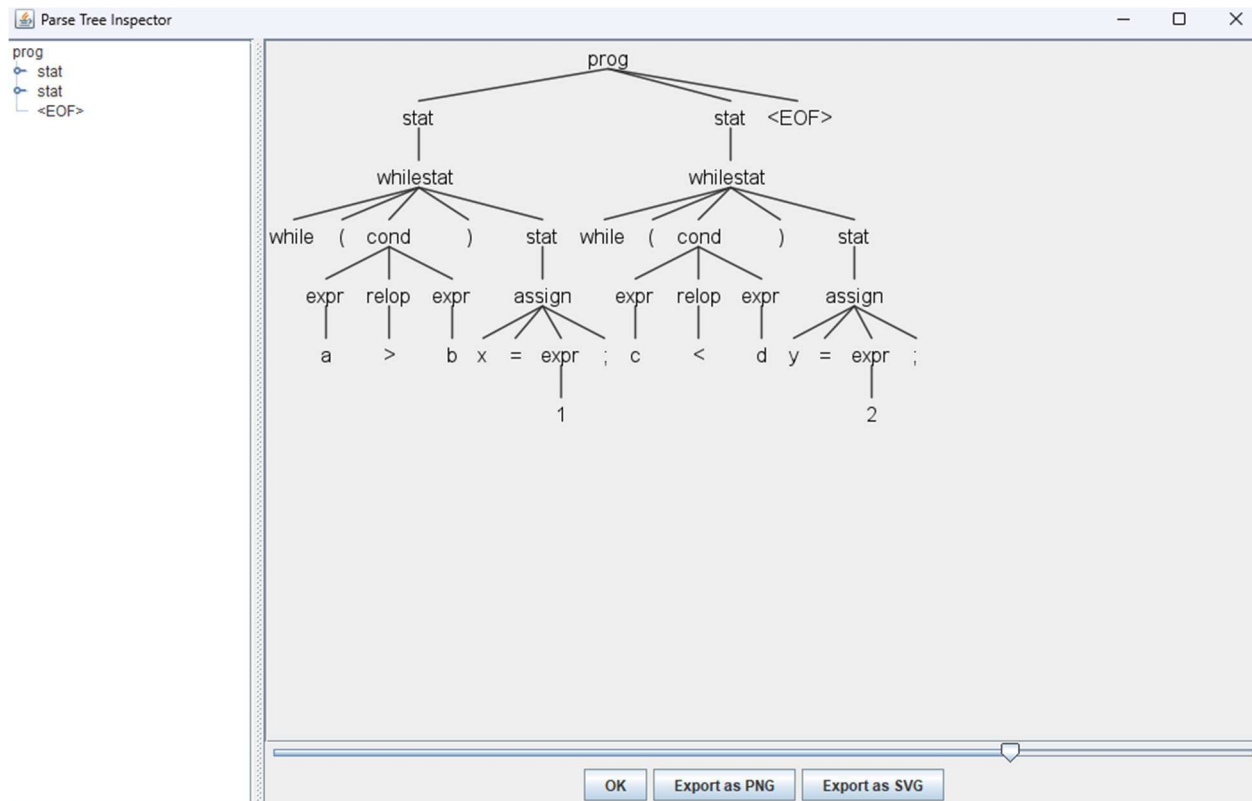
Sample Program:

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
(prog (stat (assign x = (expr 5) ;)) (stat (whilestat while ( (cond (expr x) (relop <) (expr 10)) ) (stat (block {
(stat (assign x = (expr (expr x) + (expr 1)) ;)) }))) <EOF>)
```



Counting the statements:

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (whilestat while ( (cond (expr a) (relop >) (expr b)) ) (stat (assign x = (expr 1) ;)))) (stat (whilestat while ( (cond (expr c) (relop <) (expr d)) ) (stat (assign y = (expr 2) ;)))) <EOF>)
Total while statements = 2
```



1. Modify the grammar to include

```

1 grammar Stmtg4;
2 prog : stat+ EOF ;
3 stat
4 : assign
5 | whilestat
6 | block
7 | expr ';'
8 ;
9 assign : ID '=' expr ';' ;
10 whilestat : 'while' '(' cond ')' stat ;
11 block : '{' stat+ '}' ;
12 cond
13 : cond '||' cond
14 | cond '&&' cond
15 | expr relop expr
16 ;
17 expr : expr ('*' | '/') expr
18 | expr ('+' | '-') expr
19 | '-' expr
20 | '!' expr
21 | expr '++'
22 | '++' expr
23 | NUMBER
24 | ID
25 | '(' expr ')'
26 ;
27 relop : '>' | '<' | '>=' | '<=' | '==' | '!=' ;
28 ID : [a-zA-Z][a-zA-Z0-9]* ;
29 NUMBER : [0-9]+ ;
30 WS : [ \t\r\n]+ -> skip ;

```

a. Logical Operators with Precedence:

a > b CC c < d

x == y || p != q

Input:

while (a > b CC c < d)

x = 5;

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (whilestat while ( (cond (cond (expr a) (relop >) (expr b)) && (cond (expr c) (relop <) (ex
pr d))) ) (stat (assign x = (expr 5) ;)))) <EOF>)
Total while statements = 1
```

b. -x, !flag

Input:

x = -y;

flag = !flag;

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (assign x = (expr - (expr y)) ;)) (stat (assign flag = (expr ! (expr flag)) ;)) <EOF>)
Total while statements = 0
```

c. i++, ++i

Input:

i++;

++i;

x = ++i;

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (expr (expr i) ++)) ;) (stat (expr ++ (expr i)) ;) (stat (assign x = (expr ++ (expr i)) ;))
<EOF>)
Total while statements = 0
```

2. Modify the grammar to include the following statements:

```
≡ Stmt.g4
1  grammar Stmt;
2  prog : stat+ EOF ;
3  stat
4  | : assign
5  | | whilestat
6  | | ifstat
7  | | forstat
8  | | switchstat
9  | | block
10 | ;
11 assign : ID '=' expr ';' ;
12 assignstmt : ID '=' expr ;
13 whilestat : 'while' '(' cond ')' stat ;
14 ifstat
15 | : 'if' '(' cond ')' stat
16 | | 'if' '(' cond ')' stat 'else' stat
17 | ;
18 forstat
19 | : 'for' '(' assignstmt ';' cond ';' assignstmt ')' stat
20 | ;
21 switchstat
22 | : 'switch' '(' ID ')' '{' casestat+ defaultstat? '}'
23 | ;
24 casestat
25 | : 'case' NUMBER ':' stat+
26 | ;
27 defaultstat
28 | : 'default' ':' stat+
29 | ;
30 block : '{' stat+ '}' ;
31 cond : expr relop expr ;
32 expr
33 | : expr ('*' | '/') expr
34 | | expr ('+' | '-') expr
35 | | NUMBER
36 | | ID
37 | | '(' expr ')'
38 | ;
39 relop : '>' | '<' | '>=' | '<=' | '==' | '!=' ;
40 ID : [a-zA-Z][a-zA-Z0-9]* ;
41 NUMBER : [0-9]+ ;
42 WS : [ \t\r\n]+ -> skip ;
43
```

a. for:for (i = 0; i < 10; i = i + 1) { x = x + i;}

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (forstat for ( (assignstmt i = (expr 0)) ; (cond (expr i) (relop <) (expr 10)) ; (assignstm
t i = (expr (expr i) + (expr 1))) ) (stat (block { (stat (assign x = (expr (expr x) + (expr i)) ;)) })))
)) <EOF>)
Total while statements = 0
```

b. simple if:if (x > 5) y = 10;

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (ifstat if ( (cond (expr x) (relop >) (expr 5)) ) (stat (assign y = (expr 10) ;)))) <EOF>)
Total while statements = 0
```

c. if with block if (x > 5) { y = 10; z = 20; }

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (ifstat if ( (cond (expr x) (relop >) (expr 5)) ) (stat (block { (stat (assign y = (expr 10
) ;)) (stat (assign z = (expr 20) ;)) })))) <EOF>)
Total while statements = 0
```

d. If-else statement: if (x > 5) y = 10; else y = 20;

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (ifstat if ( (cond (expr x) (relop >) (expr 5)) ) (stat (assign y = (expr 10) ;)) else (sta
t (assign y = (expr 20) ;)))) <EOF>)
Total while statements = 0
```

e. Nested if and nested if-else statements.

Input: if (x > 5){if (y > 10)z = 20;else z = 30;}

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar
TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (ifstat if ( (cond (expr x) (relop >) (expr 5)) ) (stat (block { (stat (ifstat if ( (cond (
expr y) (relop >) (expr 10)) ) (stat (assign z = (expr 20) ;)) else (stat (assign z = (expr 30) ;)))) }
)))) <EOF>)
Total while statements = 0
```


f. Switch statements

Input: switch (x) {case 1: y = 10; case 2: y = 20; default: y = 0;}

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (switchstat switch ( x ) { (casestat case 1 : (stat (assign y = (expr 10) ;))) (casestat case 2 : (stat (assign y = (expr 20) ;))) (defaultstat default : (stat (assign y = (expr 0) ;))) }))) <EOF>)
Total while statements = 0
```

3. Count the number of statements including:

- total while statements.
- only nested while statements.
- Nested if statements.
- Nested for statements.

Input:

```
while (a > b)
{
    while (c > d)
        x = 1;
}

if (x > 5)
{
    if (y > 10)
        z = 20;
}

for (i = 0; i < 5; i = i + 1)
{
    for (j = 0; j < 3; j = j + 1)
        x = x + 1;
}
```

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\Labsheet 5>java -cp .;antlr4-4.9.2-complete.jar TestStmt
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Parse Tree:
(prog (stat (whilestat while ( (cond (expr a) (relop >) (expr b)) ) (stat (block { (stat (whilestat while ( (cond (expr c) (relop >) (expr d)) ) (stat (assign x = (expr 1) ;))) }))) (stat (ifstat if ( (cond (expr x) (relop >) (expr 5)) ) (stat (block { (stat (ifstat if ( (cond (expr y) (relop >) (expr 10)) ) (stat (assign z = (expr 20) ;))) }))) (stat (forstat for ( (assignstmt i = (expr 0)) ; (cond (expr i) (relop <) (expr 5)) ; (assignstmt i = (expr (expr i) + (expr 1))) ) (stat (block { (stat (forstat for ( (assignstmt j = (expr 0)) ; (cond (expr j) (relop <) (expr 3)) ; (assignstmt j = (expr (expr j) + (expr 1))) ) (stat (assign x = (expr (expr x) + (expr 1)) ;))) }))) ) <EOF>)
Total while statements = 2
Nested while statements = 1
Nested if statements = 1
Nested for statements = 1
```